IE7275 - Data Mining in Engineering

Project 1: Traffic Crash Analysis: Chicago Dataset

Group 2

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1. Introduction

Traffic safety is a critical public concern, with road crashes affecting millions of individuals each year across the United States. Crashes result not only in physical injuries and fatalities but also in significant economic burden through healthcare costs, lost productivity, and infrastructure damage. Despite the availability of large-scale crash datasets, the factors contributing to injury severity and crash frequency are often underutilized in informing policy and preventive measures.

This project seeks to use data to enhance our understanding of traffic crash patterns, identify key risk factors that contribute to injuries and fatalities, and offer predictive insights. Through a detailed analysis of the Chicago Traffic Crashes dataset, we aim to uncover trends related to time of day, weather conditions, speed limits, and crash types that could improve how city planners, law enforcement, and public health officials approach road safety. By leveraging data analytics and machine learning, we aim to build an informed model to predict injury risk based on environmental, temporal, and situational factors, ultimately working toward more effective prevention strategies.

2. Background

Road traffic injuries are among the leading causes of death and disability worldwide. In the United States alone, the Centers for Disease Control and Prevention (CDC) reports that motor vehicle crashes are a leading cause of injury-related deaths. Urban environments like Chicago present complex traffic dynamics involving pedestrians, cyclists, and motor vehicles sharing roads with varying speed limits, traffic control devices, and infrastructure quality.

Despite the increasing availability of open traffic data through city portals, comprehensive data-driven analyses of crash patterns remain limited in their application to public safety planning. Many existing traffic safety approaches rely on aggregate statistics rather than granular, record-level analysis. With large datasets now accessible through portals like the Chicago Data Portal, there is an opportunity to move toward more targeted, evidence-based traffic safety interventions. This project explores how crash-level data can be used to identify risk patterns and predict injury outcomes across different conditions.

3. Objectives

This project has several key objectives that will help in understanding traffic crash patterns and improving tools for prediction and prevention. The first objective is to prepare the data, which involves cleaning the dataset by handling missing values, removing irrelevant columns, correcting data inconsistencies, and converting variables into suitable formats for analysis.

The next objective is to analyze the data to understand how different factors — such as time of day, weather condition, speed limit, and crash type — influence the likelihood of injury. This involves exploratory data analysis (EDA) techniques to uncover trends and correlations within the data. Following this, the project focuses on identifying patterns and insights that provide a better understanding of high-risk crash scenarios.

Evaluating the results is another key objective, where statistical metrics such as injury probability, confidence intervals, z-tests, and correlation coefficients are used to assess relationships between variables. Finally, the project focuses on visualizing the results through histograms, bar charts, scatter plots, time series plots, and ROC curves to present findings in a clear and interpretable way, supporting data-driven decisions in traffic safety planning.

Task 1: Dataset Selection/Collection

1.1 Data Collection

The analysis uses crash-level data gathered from the Chicago Traffic Crashes – Crashes dataset, publicly available through the Chicago Data Portal (data.gov). This dataset contains detailed records of individual traffic crashes reported by the Chicago Police Department (CPD), covering factors such as crash date and time, weather condition, lighting condition, road surface, traffic control devices, crash type, posted speed limit, and injury outcomes. It provides a comprehensive view of traffic crash trends across different temporal and environmental conditions, enabling analysis of the factors that impact crash severity.

The dataset contains 1,029,797 records across 48 columns, spanning crash events from 2013 through 2025. The dataset can be accessed here: https://catalog.data.gov/dataset/traffic-crashes-crashes.

Task 2: Data Acquisition and Inspection

2.1 Data Acquisition

The dataset used for this analysis was obtained from the Chicago Traffic Crashes – Crashes dataset available through the Chicago Data Portal (data.gov). The data was downloaded in CSV format and loaded into a Python environment using the Pandas library for further analysis.

The dataset contains 1,029,797 records and 48 columns, representing individual crash events reported by the Chicago Police Department. The dataset spans crash occurrences from 2013 through 2025, making it suitable for both temporal trend analysis and crash severity studies.

The dataset was loaded using the following approach:
- Imported the Pandas library
- Read the CSV file into a DataFrame
- Verified successful loading by displaying the first few rows

The resulting DataFrame occupies approximately 377 MB of memory, reflecting the large scale of the data.

2.2 Data Inspection

After loading the dataset into a Pandas DataFrame, an initial inspection was performed to understand its size, structure, variable types, and data completeness. The dataset contains **1,029,797 crash records** with **48 columns**, indicating a large-scale, crash-level dataset suitable for detailed analysis. Viewing the **first and last five rows** confirmed that each row represents a single reported crash event and includes a mix of temporal, environmental, road/traffic, injury, and location-related fields.

From the DataFrame information summary, the dataset includes **31 categorical (string) variables**, **6 integer variables**, and **11 float variables**. Key analysis-ready variables include crash timing fields such as **CRASH_DATE, CRASH_HOUR, CRASH_DAY_OF_WEEK, and CRASH_MONTH**, roadway and traffic context such as **POSTED_SPEED_LIMIT** and **TRAFFIC_CONTROL_DEVICE**, and outcome indicators such as **MOST_SEVERE_INJURY** and the injury count fields (**INJURIES_TOTAL, INJURIES_FATAL, INJURIES_INCAPACITATING**, etc.). Geographic fields (**LATITUDE, LONGITUDE, LOCATION**) are also present, enabling spatial analysis.

Missing value checks showed that most core variables needed for severity and condition-based analysis are largely complete. Fields such as **CRASH_DATE, POSTED_SPEED_LIMIT, WEATHER_CONDITION, LIGHTING_CONDITION, CRASH_TYPE**, and **PRIM_CONTRIBUTORY_CAUSE** contain little to no missing data, and injury-related columns have only a small fraction of missing values (around **0.2%**). In contrast, some fields are highly sparse and not reliable for broad analysis, including **WORKERS_PRESENT_I, WORK_ZONE_TYPE, WORK_ZONE_I, DOORING_I, PHOTOS_TAKEN_I**, and **LANE_CNT**, which have very high missing percentages. For this reason, the analysis prioritizes the well-populated crash context and outcome variables.

A statistical summary of numeric columns shows that the **posted speed limit averages ~28 mph**, and crashes typically involve about **2 units (vehicles)**. Injury counts are generally low per crash, with the average **total injuries ~0.2**, and fatal injuries being rare. Crash time fields span all hours, days, and months, supporting temporal trend analysis. Overall, the dataset structure and completeness indicate it is suitable for analyzing how environmental and roadway conditions relate to crash outcomes and severity.

Task 3: Data Cleaning and Preparation

3.1 Handling Missing Data

The dataset initially contained 48 columns, several of which had extremely high missing percentages. Columns with more than 95% missing values — including WORKERS_PRESENT_I, DOORING_I, WORK_ZONE_TYPE, WORK_ZONE_I, PHOTOS_TAKEN_I, STATEMENTS_TAKEN_I, and NOT_RIGHT_OF_WAY_I — were removed, as they did not contain sufficient usable information. Additionally, CRASH_DATE_EST_I (92.7% missing) and LANE_CNT (80.7% missing with unrealistic values) were dropped to prevent noise and instability in analysis.
For injury-related fields, missing values were minimal (≈0.2%). Since these variables represent counts, missing entries were treated as zero injuries to maintain logical consistency. Categorical variables with small missing percentages, such as REPORT_TYPE and MOST_SEVERE_INJURY, were filled with "UNKNOWN." Minor missing values in street-related fields were also replaced with "UNKNOWN." Geographic coordinates (LATITUDE, LONGITUDE, LOCATION) were left as missing rather than artificially filled.
After cleaning, missing values were reduced substantially, with only about 0.76% missing in geographic coordinates and 0.76% in posted speed limit remaining.

3.2 Handling Duplicates and Inconsistencies

The dataset was checked for duplicate crash records using CRASH_RECORD_ID, which is expected to be unique for each crash. No duplicate identifiers were found, confirming record-level integrity.
Numeric inconsistencies were addressed by converting relevant fields to numeric format using safe coercion. For example, POSTED_SPEED_LIMIT values less than or equal to zero were treated as invalid and converted to missing, as they typically indicate unknown entries. Unrealistic extreme values (such as abnormal lane counts) were handled by removing unreliable columns during preprocessing.

3.3 Data Type Conversion and Normalization

To ensure proper analysis, multiple fields were converted to appropriate data types. Date-related columns such as CRASH_DATE and DATE_POLICE_NOTIFIED were converted to datetime format. From these, derived time features such as crash year, weekday name, and week number were extracted to enable temporal analysis.
Numeric columns including POSTED_SPEED_LIMIT, LATITUDE, LONGITUDE, and BEAT_OF_OCCURRENCE were explicitly converted to numeric types. This prevented hidden formatting issues and ensured compatibility with statistical functions and visualizations.

3.4 Encoding Categorical and Indicator Variables

Binary indicator columns such as INTERSECTION_RELATED_I and HIT_AND_RUN_I were standardized and encoded into numeric format (1 for yes, 0 for no). Variations in text formatting (e.g., “Y”, “YES”, “TRUE”, etc.) were normalized before encoding to ensure consistency.
Other categorical variables were retained as string categories for grouping and visualization purposes, as they represent meaningful crash characteristics such as weather condition, lighting condition, crash type, and trafficway type.

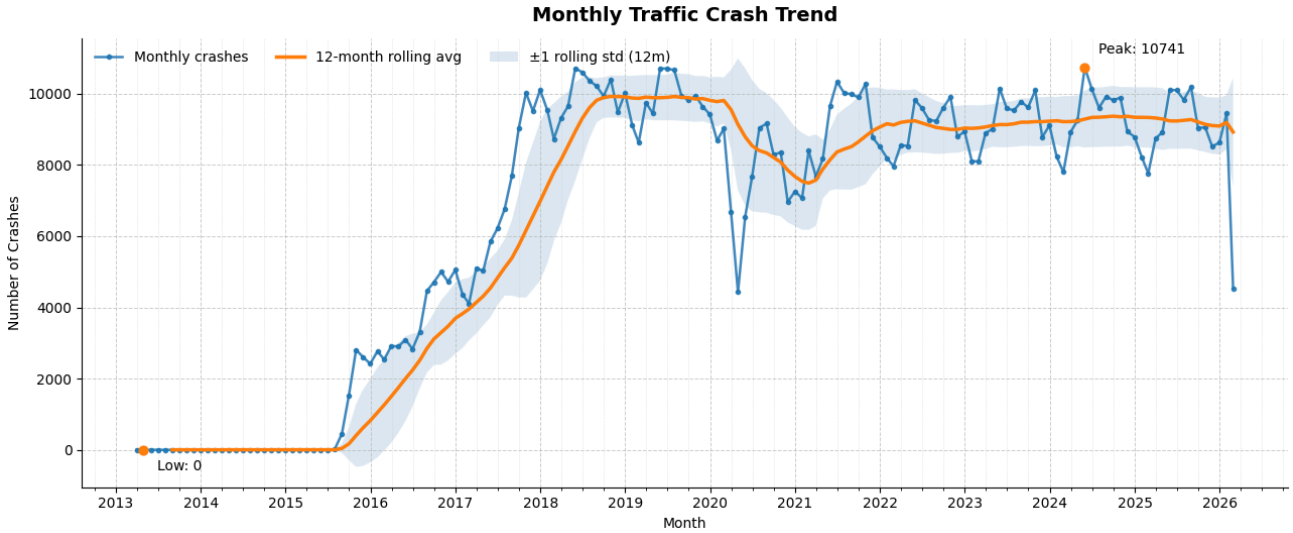
After cleaning and preparation, the dataset retained 1,029,797 records and 42 columns, with improved structural consistency, minimal missing values, and analysis-ready data types suitable for statistical modeling and visualization.

Task 4: Exploratory Data Analysis (EDA) with a Focus on Static Visualization

4.1 Monthly Crash Trends

Monthly crash counts were aggregated and analyzed using a line chart supported by a 12-month rolling mean and standard deviation. The average number of crashes per month is approximately 6,601, with a standard deviation of 3,838, reflecting substantial month-to-month variability.

The time-series visualization shows a clear upward trend beginning around 2016, stabilizing at roughly 9,000–10,000 crashes per month in later years. A temporary decline appears around 2020–2021, likely due to reduced travel patterns. The highest monthly crash count observed was **10,741**. Rolling averages confirm long-term structural trends by smoothing short-term fluctuations.



4.2 Crash Frequency by Hour of the Day

A bar plot of crashes across the 24-hour cycle (Fig 4.2.1) shows that incidents rise sharply during morning commute hours, stay elevated through midday, and peak at **15:00 (3 PM)** with **80,013 crashes**. The lowest crash frequencies occur in early morning hours (2–5 AM). This distribution closely aligns with traffic density and commuting behavior.

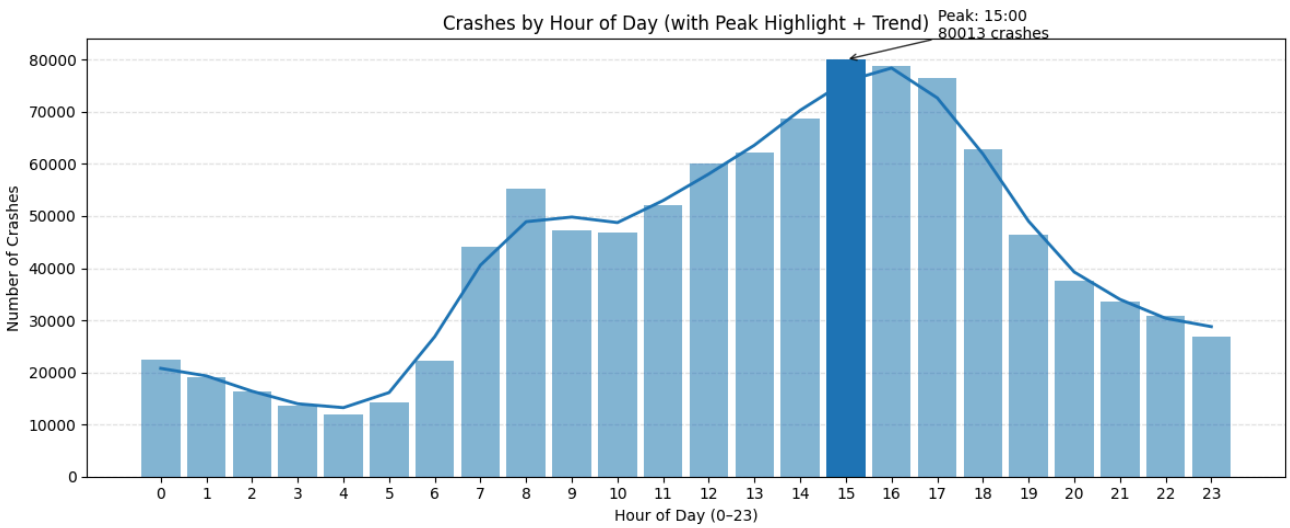


Fig 4.2.1

4.3 Overall Injury and Fatality Probabilities

Binary indicators were used to estimate population-level probabilities:

Injury probability: 0.1432 (14.32%)
Fatal crash probability: 0.001056 (0.106%)

This means approximately one out of seven crashes results in at least one injury, while fatal crashes remain extremely rare. A 95% confidence interval for the injury probability was calculated as **(0.1425, 0.1438)**, indicating high statistical precision due to the large dataset.

4.4 Weekday vs. Weekend Injury Risk

Conditional probabilities were compared for weekend and weekday crashes:

P(Injury | Weekend): 0.1485
P(Injury | Weekday): 0.1411
Difference: 0.0074
A z-test for the difference in proportions yielded **Z = 9.532**, which exceeds the 5% significance threshold (|Z| > 1.96). Although the numerical difference is small, it is statistically significant, indicating slightly higher injury risk during weekends.

4.5 Weather Conditions and Injury Probability

Injury probabilities were computed for weather categories with sufficient data (≥ 500 cases):

Blowing snow: ~19.2%
Fog/Smoke/Haze: ~18.2%
Rain: ~17.5%
Clear: ~14.7%
Snow: ~12.3%
Unknown: ~5.3%

Adverse weather and low-visibility conditions significantly increase the chance of injury (Fig 4.5.1). Clear weather exhibits moderate injury probability, largely due to higher overall exposure.

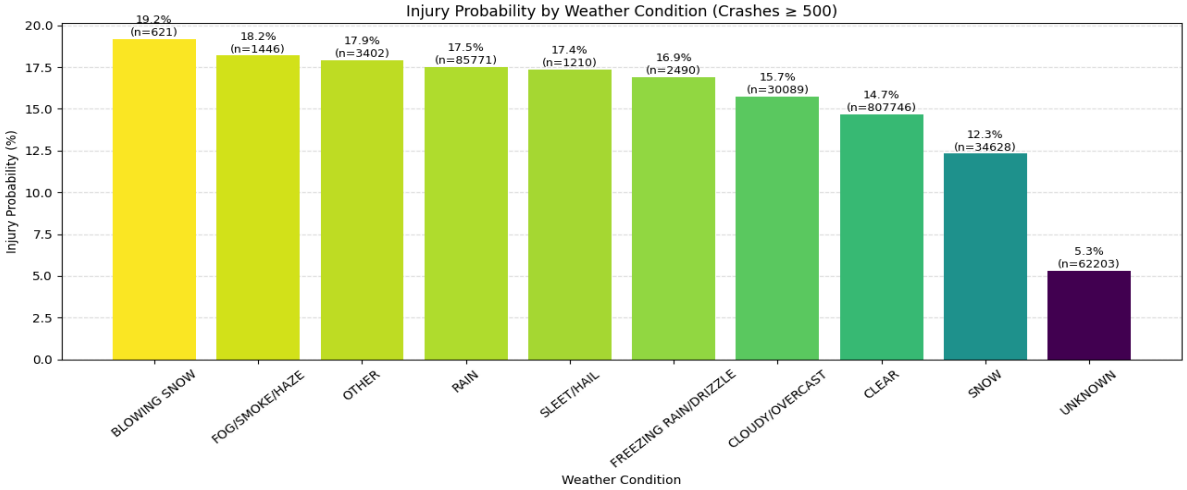


Fig 4.5.1

4.6 Relationship Between Speed Limit and Injury Risk

A scatter plot and correlation analysis (Fig 4.6.1) were used to evaluate how speed limits relate to injury probability.

Aggregated (binned) correlation: ~0.922
Record-level correlation: ~0.0807

This indicates a strong positive relationship when speed limits are analyzed in grouped ranges, but only a weak correlation at the individual crash level. Thus, speed contributes to injury risk but does not act as a standalone predictor.

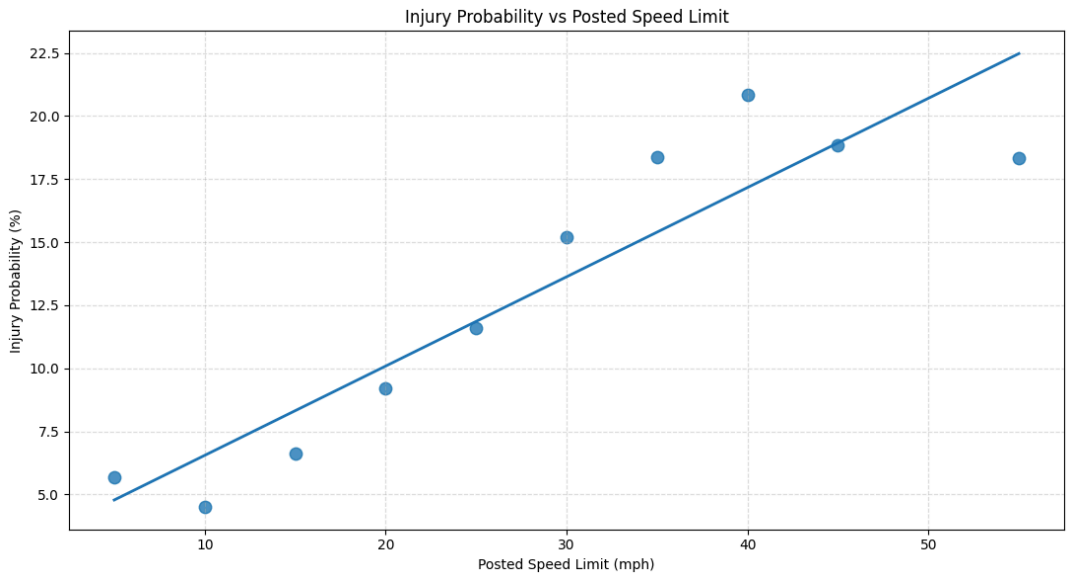


Fig 4.6.1

4.7 Primary Contributing Causes of Crashes

A horizontal bar chart (Fig 4.7.1) of the top contributing factors shows:

1) Unable to Determine – 405,372
2) Failure to Yield Right-of-Way – 114,671
3) Following Too Closely – 98,509
4) Improper Passing/Overtaking – 51,892
5) Failure to Reduce Speed – 43,151

Driver behavior is the most dominant influence on crash causation. The large "Unable to Determine" category suggests inconsistency or limitations in reporting.

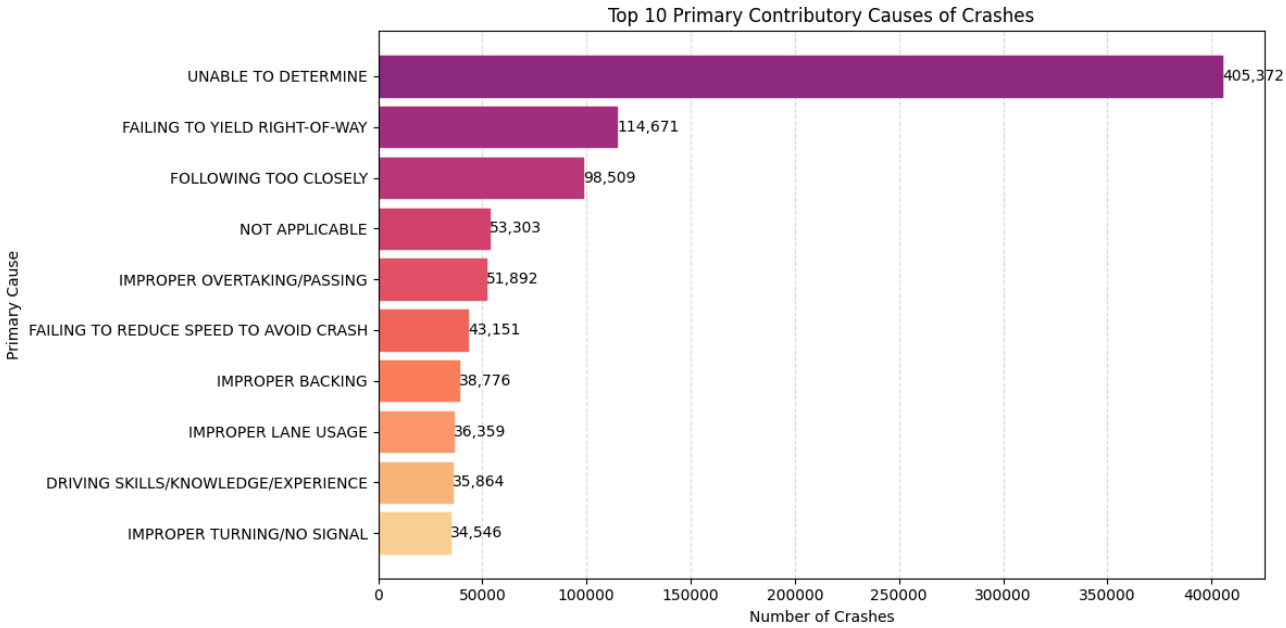


Fig 4.7.1

4.8 Distribution of Injury Counts (Skewness and Spread)

Key descriptive statistics for injuries per crash:

1) **Mean:** 0.1977
2) **Median:** 0
3) **Variance:** 0.3333
4) **Skewness:** 4.6979

Percentiles show:

1) 50% → 0
2) 90% → 1
3) 95% → 1
4) 99% → 3

The distribution is **highly right-skewed**, meaning most crashes involve no injuries, while a small number involve multiple injuries. This explains the low overall average despite occasional severe events.

Task 5: Advanced Analysis

5.1. Data Understanding and Preprocessing

The project analyzes a large multi-year crash dataset with rich temporal, roadway, vehicle, and injury attributes. After importing and inspecting the raw file, extensive cleaning was performed: invalid or placeholder codes were standardized, key numeric fields were cast to appropriate types, and records with structurally impossible values were removed to ensure internal consistency.

Missing values were handled using a mixed imputation strategy: median imputation for continuous variables (e.g., posted speed, crash hour, number of units) and most-frequent (mode) imputation for categorical variables. This approach preserves the empirical distribution of the data while avoiding the sample size reduction that would result from listwise deletion, which is crucial given the downstream use in supervised learning.

For modeling, the dataset was split into numerical and categorical feature blocks. Numerical variables were scaled and imputed, while categorical variables were transformed via one-hot encoding with "ignore unknown" behavior to maintain robustness on new categories at prediction time. The final modeling matrix thus combines standardized continuous features with a high-dimensional sparse set of binary indicators representing discrete conditions at crash time.

5.2. Baseline Injury Risk and Time Patterns

The unconditional probability that a crash involves at least one injury is about 14.3%, while the probability of a fatal crash is about 0.11%. This confirms that most reported crashes are property-damage-only, with a small but important fraction resulting in injuries or fatalities.

Crash frequency by hour of day shows a clear pattern: the peak crash hour is 15:00 (3 pm), with higher counts during afternoon "commute" hours and lower counts overnight. A smoothed hourly trend confirms that afternoon traffic exposure and congestion are key drivers of when crashes occur.

Comparing injury probability by day type, weekend crashes have an injury probability of about 0.1485 versus 0.1411 on weekdays. The difference is small but stable, suggesting slightly higher conditional injury risk on weekends, possibly due to different trip purposes and behavior

5.3. Speed and Injury Probability

When injury probability is averaged by posted speed limit category, higher speed limits show higher mean injury probabilities, with a correlation around 0.922 between speed category and injury probability. This indicates that crashes on higher-speed roads are more likely to cause injuries once a crash happens. However, the overall micro-level correlation between numeric posted speed and the injury indicator across all records is only about 0.0807. This contrast shows that simple pairwise correlation on raw data can hide important patterns that become clear after grouping by relevant categories.

5.4. Logistic Regression Injury Model

A logistic regression classifier was built to predict whether a crash involves an injury. Numeric predictors include posted speed limit, crash hour, crash month, number of units, and a weekend indicator; categorical predictors include weather, lighting, traffic control, surface condition, trafficway type, and primary crash type, all one-hot encoded within a scikit-learn pipeline.

The model is trained on a stratified train–test split and evaluated using ROC–AUC and standard classification metrics. The ROC–AUC is about 0.78–0.79, showing reasonably strong discrimination between injury and non-injury crashes given the available variables.

5.5. Key Predictors of Injury

Mapping the logistic regression coefficients back to feature names identifies the strongest factors affecting injury odds. The largest positive coefficients (higher injury risk) are:

FIRSTCRASHTYPE = PEDESTRIAN (coef ≈ +3.41)
FIRSTCRASHTYPE = PEDALCYCLIST (≈ +2.26)
NUMUNITS (per extra unit, ≈ +0.95)
FIRSTCRASHTYPE = OVERTURNED (≈ +0.74)
FIRSTCRASHTYPE = FIXED OBJECT and other severe crash types.

These indicate much higher injury odds for crashes involving vulnerable users (pedestrians, cyclists), rollovers, fixed-object impacts, head-on collisions, and multi-vehicle crashes.

The most negative coefficients (lower injury risk) include:

FIRSTCRASHTYPE = PARKED MOTOR VEHICLE (≈ −2.04)
Several rear-end and sideswipe crash types.

TRAFFICWAYTYPE = ALLEY, PARKING LOT, DRIVEWAY, and some “UNKNOWN” low-speed contexts.

These conditions typically involve lower speeds or glancing impacts, so injury is less likely. Some “UNKNOWN” categories show negative coefficients, likely reflecting under-reporting of details in low-severity crashes rather than true protective effects.

Overall, the model shows that injury risk depends strongly on the physical crash configuration, with environment and time-of-day playing secondary roles.

6. Main Insights and Limitations

The analysis supports three main insights:

1. Crashes with pedestrians, cyclists, rollovers, fixed objects, and multi-vehicle involvement have much higher injury probabilities.
2. Higher posted speed environments are associated with higher conditional injury risk, even though raw speed–injury correlation at record level is weak.
3. Peak crash occurrence is in the afternoon, and weekend crashes show slightly higher injury risk than weekday crashes.

Key limitations are that the model is observational (associational, not causal), assumes linear log-odds effects, and uses reported data that may be biased, especially for minor crashes. Future work could add non-linear models, spatial and exposure variables, and formal statistical tests for specific hypotheses about individual risk factors

6. Conclusion

In this project, crash injury risk is shown to depend mainly on crash type, involvement of vulnerable users, and road speed environment.

Crashes involving pedestrians, cyclists, rollovers, fixed objects, and multi-vehicle involvement have much higher injury odds than low-speed or parking-related crashes. Higher posted speed limits are associated with higher average injury probability when crashes are grouped by speed category.

Crashes occur most frequently in the afternoon (around 3 pm), and weekend crashes have a slightly higher probability of injury than weekday crashes. A logistic regression model (ROC–AUC ≈ 0.78–0.79) confirms these patterns and highlights severe crash types, higher speeds, and vulnerable road users as the main contributors to injury risk in this dataset.